



RV College of Engineering®
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NBA Accredited (UG - 6 Years
PG - 3 Years)

Department of Electronics & Communication Engineering

CIE-1

Course Code: EC123ATC

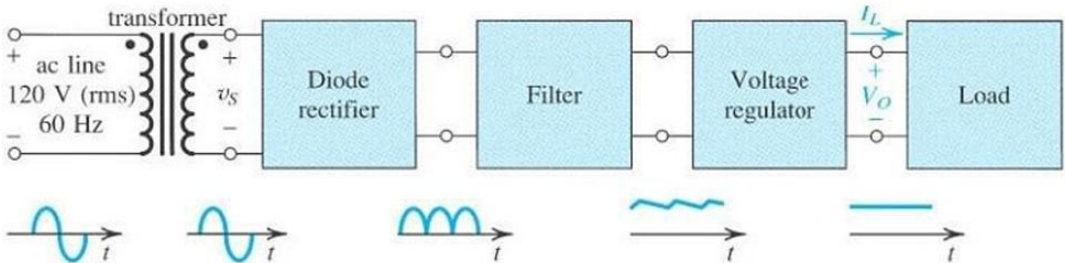
Course Title: Principles of Electronics Engineering

Semester: II

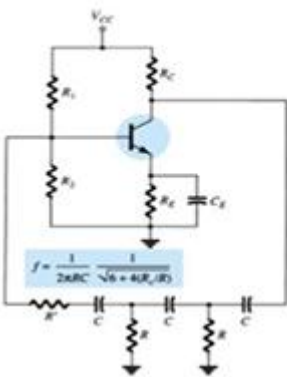
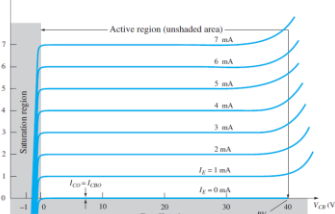
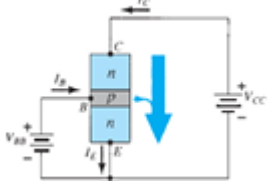
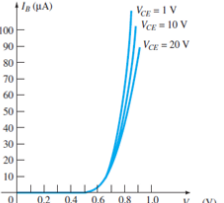
Slno	Quiz	Marks
1	Avalanche	1M
2	Forward	1M
3	28	1M
4	180	1M
5	499	1M
6	0.99	1M
7	VBE, I _B	1M
8	39.54	1M
9	ripple	1M
10	2	1M

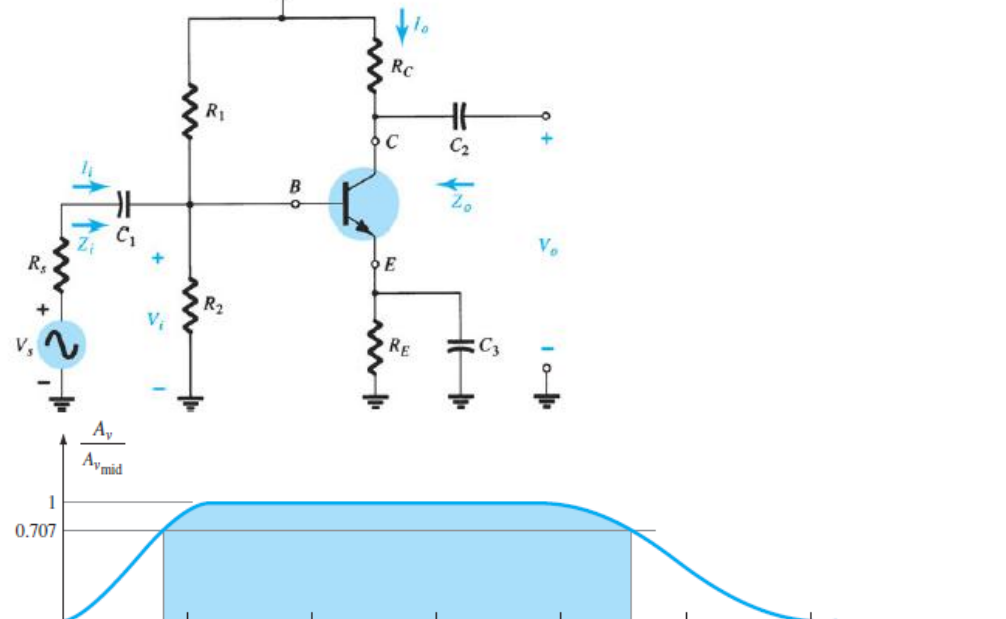


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Sl. No.	Solution with Mark split-up	Marks
1.a	 Explanation of each block	2 4
1.b	$R_L = 144.34 \Omega$ $V_{dc} = 72.17 V$ $\% \text{ load regulation} = 3.64\%$	2 1 1
2a	$I_{z(\max)} = P_{d(\max)} / V_Z = 0.194 A$ $I_{L(\max)} = V_Z / R_{L(\min)} = 60 \text{ mA}$ $I_{L(\min)} = V_Z / R_{L(\max)} = 6 \text{ mA}$ When $V_{in} = 12 V$ $[(V_{in(\min)} - V_Z) / R] - I_{L(\max)} > I_{Z(\min)}$ $R < 60 \Omega$ When $V_{in} = 18 V$ $[(V_{in(\max)} - V_Z) / R] - I_{L(\min)} < I_{z(\max)}$ $R > 90.9 \Omega$ $60 \Omega < R < 90.9 \Omega$	3 3
2.b	PHOTO DIODE A photodiode is a kind of light detector, which involves the conversion of light into voltage or current, based on the mode of operation of the device. Working Principle of Photodiodes When photons of energy greater than 1.1 eV hit the diode, electron-hole pairs are created. The intensity of photon absorption depends on the energy of photons – the lower the energy of photons, the deeper the absorption is. This process is known as the inner photoelectric effect. Applications Photodiodes find application in the following: Cameras, Medical devices Safety equipment, Optical communication devices Position sensors	2 1 1
3.a	$A_f = 25$ $Z_{of} = 25 \Omega$ $Z_{if} = 600 K \Omega$ $1 + A\beta = 40$ $BW_f = 1.2 \text{ MHz}$ $D_f = 0.375\%$	1 1 1 1 1 1

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3.b	 Working Frequency of oscillation $f = \frac{1}{2\pi RC\sqrt{6}}$ $\beta = \frac{1}{29}$ $A > 29$	4
4.a	The 3 main regions of operation of a transistor are:- 1. Cutoff 2. Saturation 3. Active region Cutoff region: Here the two junctions EB and CB are reverse biased. In this region the device behaves as an open switch. Saturation region: Both the junctions are forward biased. In this region the device acts like a closed switch. Active region: Here the EB junction is forward biased and CB junction is reverse biased. In this region the BJT is used as an amplifier. Note: When EB junction is reverse biased and CB junction is forward biased, this region is called as inverse active region. The operation in this region is generally not used in practice.   	4 2
4.b	Gain stability with feedback: $\left \frac{dA_f}{A_f} \right = \frac{1}{ 1 + \beta A } \left \frac{dA}{A} \right $ $\left \frac{dA_f}{A_f} \right \approx \left \frac{1}{\beta A} \right \left \frac{dA}{A} \right \quad \text{for } \beta A \gg 1$ This shows that magnitude of the relative change in gain $\left \frac{dA_f}{A_f} \right$ is reduced by the factor βA compared to that without feedback $\left(\left \frac{dA}{A} \right \right)$.	4

5.a		
	 The circuit diagram shows a common-emitter amplifier. The input signal V_s is connected to the base through a series resistor R_s and a coupling capacitor C_1. The base is biased by a voltage divider consisting of R_1 and R_2 connected to V_{CC} and ground. The emitter is connected to ground through a resistor R_E and a bypass capacitor C_3. The collector is connected to V_{CC} through a resistor R_C and has a load capacitor C_2 connected to the output. The output voltage is V_o. The input impedance is Z_i and the output impedance is Z_o. The current gain is β. The frequency response plot shows the voltage gain A_v versus frequency f on a log scale. The gain is 1 at midband frequencies and drops to 0.707 at the lower cutoff frequency f_L and the upper cutoff frequency f_H. The midband gain is A_{vmid}.	3 3
5.b	Explanation Load Effect: ΔV_0 (no load to full load)=18-17.9=0.1V Source Effect: ΔV_0 for a 10% change in V_s=18-17.95=0.05V Load Regulation: (Load Effect/V_0)*100=(0.1/18)*100=0.5% -----2M Line Regulation: (Source Effect/V_0)*100=(0.05/18)*100=0.27%-----2M	2 2

Signature (Digital) of course handling faculty

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